Appendix: Where Common Knowledge Cannot Be Formed, Common Belief Can – Planning with Multi-Agent Belief Using Group Justified Perspectives

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A Theorem 1 Proof

The following theorem is proposed in the paper.

Theorem 1. *For any agent $i \in A$ and perspective $\vec{s} \in \vec{S}$:*

$$f_i(\vec{s}) = f_i(f_i(\vec{s}))$$

Proof. Let $s(v) = \tau$ be $v \notin s$ in the following content.

The base case is a sequence with one element $[s_0]$. In this case, the above lemma holds trivially, because: $f_i([s_0]) = [O_i(s_0)]$; $f_i(f_i([s_0])) = f_i([O_i(s_0)]) = [O_i(O(s_0))]$; and, $O_i(s_0) = O_i(O(s_0))$.

For a sequence for more than one element, $f_i(\vec{s})[t]$ depends on the $[O_i(s_0), \ldots, O_i(s_t)]$ and $s_0, \ldots, s_t$. Let v be any variable in V.

If we have $v \in O_i(s_t)$, then, we have $f_i(\vec{s})[t](v) = O_i(s_t)(v) = s_t(v)$ and $f_i(f_i(\vec{s}))[t](v) = O_i(f_i(\vec{s})[t])(v) = O_i(O(s_t))(v) = O_i(s_t)(v) = s_t(v)$, since:$v \in O_i(s_t)$; $O_i(s_t) = O_i(O_i(s_t))$; and, $O_i(s_t) \subseteq f_i(\vec{s})[t] \rightarrow O_i(O_i(s_t)) \subseteq O_i(f_i(\vec{s})[t])$.

If we have $v \notin O_i(s_t)$, then, we have 2 possibilities: either $v \in O_i(f_i(\vec{s})[t])$ or $v \notin O_i(f_i(\vec{s})[t])$. If v in $O_i(f_i(\vec{s})[t])$, then, we have $f_i(f_i(\vec{s}))[t](v) = O_i(f_i(\vec{s})[t])(v) = f_i(\vec{s})[t]$. Otherwise, whether $f_i(\vec{s})[t](v) = f_i(f_i(\vec{s}))[t](v)$ holds depends on whether $f_i(\vec{s})[t-1](v) = f_i(f_i(\vec{s}))[t-1](v)$. When $t-1$ is 0, it is our base case above.

Overall, $f_i(\vec{s}) = f_i(f_i(\vec{s}))$ holds. □

B Complete Semantics

Here, we provide the full complete semantics, which includes: semantics for the individual modal logic operators using justified perspective functions (item a-g) [Hu *et al.*, 2023]; semantics for the group seeing and knowledge operators using group perspective functions (item h-l) [Hu *et al.*, 2022]; And, the semantics for group belief operators we proposed in Section 3.2.

Definition B .1 (Complete semantics)**.** The complete semantics for justified perspective are defined as:

(a)	$(M, \vec{s}) \vDash r(\vec{t})$	iff	$\pi(s_n, r(\vec{t})) = true$
(b)	$(M, \vec{s}) \vDash \phi \wedge \psi$	iff	$(M, \vec{s}) \vDash \phi$ and $(M, s) \vDash \psi$
(c)	$(M, \vec{s}) \vDash \neg\varphi$	iff	$(M, \vec{s}) \nvDash \varphi$
(d)	$(M, \vec{s}) \vDash S_i v$	iff	$v \in \mathrm{dom}(O_i(s_n))$
(e)	$(M, \vec{s}) \vDash S_i\varphi$	iff	$\forall \vec{g} \in \vec{S}_{tG}, (M, g[O_i(s_n)]) \vDash \varphi$ or $\forall \vec{g} \in \vec{S}_{tG}, (M, g[O_i(s_n)]) \vDash \neg\varphi$
(f)	$(M, \vec{s}) \vDash K_i\varphi$	iff	$(M, \vec{s}) \vDash \varphi \wedge S_i\varphi$
(g)	$(M, \vec{s}) \vDash B_i\varphi$	iff	$\forall \vec{g} \in \vec{S}_{tG}, (M, \vec{g}[f_i(\vec{s})]) \vDash \varphi$
(h)	$(M, \vec{s}) \vDash ES_G\alpha$	iff	$\forall i \in G, (M, \vec{s}) \vDash S_i\alpha$
(i)	$(M, \vec{s}) \vDash DS_G v$	iff	$v \in \mathrm{dom}(\bigcup_{i \in G} O_i(s_n))$
(j)	$(M, \vec{s}) \vDash DS_G\varphi$	iff	$\forall g \in \mathcal{S}_G, (M, g[s']) \vDash \varphi$ or $\forall g \in \mathcal{S}_G, (M, g[s']) \vDash \neg\varphi$, where $s' = \bigcup_{i \in G} O_i(s_n)$
(k)	$(M, \vec{s}) \vDash CS_G v$	iff	$v \in \mathrm{dom}(cO(G, s_n))$
(l)	$(M, \vec{s}) \vDash CS_G\varphi$	iff	$\forall g \in \mathcal{S}_G, (M, g[s']) \vDash \varphi$ or $\forall g \in \mathcal{S}_G, (M, g[s']) \vDash \neg\varphi$, where $s' = cO(G, s_n)$
(m)	$(M, \vec{s}) \vDash EB_G\varphi$	iff	$\forall \vec{w} \in ef_i(\vec{s}), \forall \vec{g} \in \vec{S}_{tG}, (M, \vec{g}[\vec{w}]) \vDash \varphi$
(n)	$(M, \vec{s}) \vDash DB_G\varphi$	iff	$\exists \vec{w} \in df_G(\vec{s}), \forall \vec{g} \in \vec{S}_{tG}, (M, \vec{g}[\vec{w}]) \vDash \varphi$
(n*)	$(M, \vec{s}) \vDash DB_G\varphi$	iff	$\forall \vec{w} \in df_G(\vec{s}), \forall \vec{g} \in \vec{S}_{tG}, (M, \vec{g}[\vec{w}]) \vDash \varphi$
(o)	$(M, \vec{s}) \vDash CB_G\varphi$	iff	$\forall \vec{w} \in cf_G(\{\vec{s}\}), \forall \vec{g} \in \vec{S}_{tG}, (M, \vec{g}[\vec{w}]) \vDash \varphi$

where: α is a variable v or a formulaφ; $\vec{S}_{tG} \subsetneq \vec{S}$ is the set of all possible global states sequences with the length of t and $\vec{g}[\vec{s}] = g_1[s_1], \ldots, g_n[s_n]$; $cO(G, s)$ is the state reached by applying composite function $\bigcap_{i \in G} O_i$ until it reaches its fixed point. That is, the fixed point s' such that $cO(G, s') = cO(G, \bigcap_{i \in G} O_i(s'))$.

C Ternary Semantics

Definition C .1 (Ternary Semantics)**.** Here we provide full ternary semantics. Similar as the complete semantics: item a-g [Hu *et al.*, 2023] are for individual modal operators using justified perspective function; item h-l [Hu *et al.*, 2022] are for group seeing and knowledge operators using group per-

spective functions; item m-o are for group belief operators defined in Section 3.2.

The ternary semantics are defined using function T, omitting the model M for readability:

$$
\begin{array}{llll}
\text{(a)} & T[\vec{s}, r(\vec{t})] & = & 1 \text{ if } \pi(s_n, r(\vec{t})) = true; \\
 & & & 0 \text{ if } \pi(s_n, r(\vec{t})) = false; \\
 & & & \frac{1}{2} \text{ otherwise} \\
\text{(b)} & T[\vec{s}, \phi \wedge \psi] & = & \min(T[\vec{s}, \phi], T[\vec{s}, \psi]) \\
\text{(c)} & T[\vec{s}, \neg\varphi] & = & 1 - T[\vec{s}, \varphi] \\
\text{(d)} & T[\vec{s}, S_i v] & = & \frac{1}{2} \text{ if } i \notin \mathrm{dom}(s_n) \text{ or } v \notin \mathrm{dom}(s_n) \\
 & & & 0 \text{ if } v \notin \mathrm{dom}(O_i(s_n)) \\
 & & & 1 \text{ otherwise} \\
\text{(e)} & T[\vec{s}, S_i\varphi] & = & \frac{1}{2} \text{ if } T[\vec{s}, \varphi] = \frac{1}{2} \text{ or } i \notin \mathrm{dom}(s_n); \\
 & & & 0 \text{ if } T[\langle O_i(s_n)\rangle, \varphi] = \frac{1}{2}; \\
 & & & 1 \text{ otherwise} \\
\text{(f)} & T[\vec{s}, K_i\varphi] & = & T[\vec{s}, \varphi \wedge S_i\varphi] \\
\text{(g)} & T[\vec{s}, B_i\varphi] & = & T[f_i(\vec{s}), \varphi] \\
\text{(h)} & T[\vec{s}, ES_G\alpha] & = & \min(\{T[s, S_i\alpha] \mid i \in G\}) \\
\text{(i)} & T[\vec{s}, DS_G v] & = & \frac{1}{2} \text{ if } v \notin \mathrm{dom}(s) \\
 & & & \text{or } \forall i \in G,\ i \notin \mathrm{dom}(s); \\
 & & & 0 \text{ if } v \notin \mathrm{dom}(\bigcup_{i\in G} O_i(s)); \\
 & & & 1 \text{ otherwise} \\
\text{(j)} & T[\vec{s}, DS_G\varphi] & = & \frac{1}{2} \text{ if } T[s, \varphi] = T[s, \neg\varphi] = \frac{1}{2} \\
 & & & \text{or } \forall i \in G,\ i \notin \mathrm{dom}(s) \\
 & & & 0 \text{ if } T[s', \varphi] = T[s', \neg\varphi] = \frac{1}{2}, \\
 & & & \text{where } s' = \bigcup_{i\in G} O_i(s); \\
 & & & 1 \text{ otherwise} \\
\text{(k)} & T[\vec{s}, CS_G v] & = & \frac{1}{2} \text{ if } v \notin \mathrm{dom}(s) \\
 & & & \text{or } \exists i \in G,\ i \notin \mathrm{dom}(s); \\
 & & & 0 \text{ if } v \notin \mathrm{dom}(cO(G, s)); \\
 & & & 1 \text{ otherwise} \\
\text{(l)} & T[\vec{s}, CS_G\varphi] & = & \frac{1}{2} \text{ if } T[s, \varphi] = T[s, \neg\varphi] = \frac{1}{2} \\
 & & & \text{or } \exists i \in G,\ i \notin \mathrm{dom}(s); \\
 & & & 0 \text{ if } T[s', \varphi] = T[s', \neg\varphi] = \frac{1}{2}, \\
 & & & \text{where } s' = cO(G, s); \\
 & & & 1 \text{ otherwise} \\
\text{(m)} & T[\vec{s}, EB_G\varphi] & = & \min(\{T[\vec{g}, \varphi] \mid \forall \vec{g} \in ef_G(\vec{s})\}) \\
\text{(n)} & T[\vec{s}, DB_G\varphi] & = & \max(\{T[\vec{g}, \varphi] \mid \forall \vec{g} \in df_G(\vec{s})\}) \\
\text{(n*)} & T[\vec{s}, DB_G\varphi] & = & \min(\{T[\vec{g}, \varphi] \mid \forall \vec{g} \in df_G(\vec{s})\}) \\
\text{(o)} & T[\vec{s}, CB_G\varphi] & = & \min(\{T[\vec{g}, \varphi] \mid \forall \vec{g} \in cf_G(\{\vec{s}\})\})
\end{array}
$$

where s_n is the final state in sequence $\vec{s}$; that is, $s_n = \vec{s}[|\vec{s}|]$.

D Problem Instances

Here we provide details on the problem instances. For easy referencing, we present the table (Table 1) as well.

D .1 Number

The problem settings, as described in Example 1, are of agents taking turns to peek into a box containing a number, where the number can be changed.

Item (N1) would be solved by just one agent peeking into the box and the number subtracted by 1. Items (N0) and (N2) are solved by Plan 1, as both of the agents a uniformly and commonly and b believe $n = 1$. Items (N3) and (N4) are more challenging and deserve some discussion.

Item (N3) means both a and b believe n is smaller than 2, while they do not commonly believe it. The returned plan is: *peek(a)*, *return(a)*, *peek(b)*, *subtraction*, *return(b)*, *peek(a)*. Intuitively, a sees n first, and then b sees n changed. When a sees n again, as the design decision made by Hu *et al.*, a will assume when b sees n, n is the value a believes (which is 2).

Item (N4) means both a and b believe that both of them believe n is smaller than 2, while they still do not commonly believe it. The returned plan is: *peek(a)*, *return(a)*, *peek(b)*, *subtraction*, *return(b)*, *peek(a)*, *return(a)*, *peek(b)*. It is simpler to show the perspectives of each agent[1]:

$$
\begin{array}{lll}
[f_a(\vec{s})] & = & [\tau, 2, 2, 2, 2, 2, 1, 1, 1] \\
[f_b(\vec{s})] & = & [\tau, \tau, 2, 2, 2, 2, 1, 1, 1] \\
[f_b(f_a(\vec{s}))] & = & [\tau, \tau, \tau, 2, 2, 2, 2, 2, 1] \\
[f_a(f_b(\vec{s}))] & = & [\tau, \tau, \tau, 2, 2, 2, 1, 1, 1] \\
[f_a(f_b(f_a(\vec{s})))] & = & [\tau, \tau, \tau, 2, 2, 2, 2, 2, 2]
\end{array}
$$

In the last state in the perspective, a sees n , where the n is $f_b(f_a(\vec{s}))[6] = 2$. So that we have $EB_G EB_G n < 2 \wedge \neg CB_G n < 2$. Items (5) and (6) are about common belief generated while a and b hold different beliefs on the value of n.

D .2 Grapevine

The problem setting is: In two adjacent rooms, 4 agents, in the same room, each has their own secret. All agents can move between two rooms and *share* or *lie* about a secret sct_i, if either the secret is their own secret, or they have heard the secret from someone. That is, they need to believe the secret ($B_i sct_i$) before they can share or lie about it. Initially, all agents a, b, c, d are located in the left room, and they only know about their own secrets.

As for the specific group beliefs, item (0) is trivial. To form a common belief in the group about sct_a, agent a just needs to share sct_a when everyone is in the same room.

Items (G1) and (G2) are more interesting. Item (G1) represents that everyone in a group believes sct_a but there is no common belief on sct_a, while item (G2) represents that everyone in a group believes that everyone believes sct_a, but still no common belief is formed. The plan to solve (G1) is still intuitive: *move-right(b)*, *sharing(a,*sct_a*)*, *move-right(a)* and *sharing(a,*sct_a*)*. One of the agents moves to the other room first, then a shares sct_a, and moves to the other room to share with that agent individually. The plan for (G2) is the same as (G1) but with 2 extra actions: *move-left(b)* and *sharing(b,*sct_a*)*. Intuitively, after everyone knows sct_a, agent b returns to the original room and tells others that it believes the sct_a as well by sharing sct_a.

Items (G3) and (G4) form different common beliefs. Item (G3) is formed by agent a lying about sct_a while b is in the room and shares the actual secret value after b leaves. While item (G4) is formed by agent a doing the same process for both agent b and c.

[1]Here, we also omit $peeking_a$ and $peeking_b$ for readability.

ID	Exp	Gen	Common		External		Total	$\|p\|$	Goals
			Max	Avg	$\|calls\|$	Avg Time(ms)	Time(s)		
N0	39	140	0	0	207	0.085	0.048	4	$EB_G n < 2$
N1	7	25	0	0	33	0.095	0.006	2	$DB_G n < 2$
N2	39	140	4	2.199	207	0.255	0.075	4	$CB_G n < 2$
N3	120	435	4	2.461	668	0.414	0.354	6	$EB_G n < 2 \wedge \neg CB_G n < 2$
N4	347	1273	5	2.716	2041	0.798	1.906	8	$EB_G EB_G n < 2 \wedge \neg CB_G n < 2$
N5	31	111	3	2.134	161	0.307	0.068	4	$\neg EB_G n = 1 \wedge \neg EB_G n = 2 \wedge CB_G n < 2$
N6	50	177	3	1.649	257	0.301	0.107	4	$B_a CB_G n = 2 \wedge B_b CB_G n = 1$
G0	5	35	4	3.028	41	3.17	0.149	1	$CB_G sct_a = t$
G1	66	450	4	3.195	691	5.94	4.542	4	$EB_G sct_a = t \wedge CB_G sct_a = t \rightarrow \frac{1}{2}$
G2	240	1828	5	3.496	3282	9.984	35.764	6	$EB_G EB_G sct_a = t \wedge CB_G sct_a = t \rightarrow \frac{1}{2}$
G3	103	913	4	3.018	1138	4.882	6.14	3	$B_b CB_G sct_a = f \wedge CB_{\{a,c,d\}} sct_a = t$
G4	328	2959	4	2.664	3792	10.717	42.776	4	$CB_{\{b,c\}} CB_G sct_a = f \wedge CB_{\{a,d\}} sct_a = t$
G5	66	450	4	3.195	691	6.469	4.89	4	$DB_G EB_G sct_a = t \wedge CB_G sct_a = t \rightarrow \frac{1}{2}$
G6	70	455	0	0	734	1.326	1.396	4	$DB_G EB_G sct_a = t \wedge B_a EB_G sct_a = t \rightarrow \frac{1}{2}$
BBL0	2	8	4	3.111	11	9.387	0.107	1	$CB_G o_2 = 2$
BBL1	5	19	4	3.143	26	10.617	0.279	2	$EB_G o_2 = 2 \wedge CB_G o_2 = 2 \rightarrow \frac{1}{2}$
BBL2	177	708	4	3.492	1011	21.724	22.072	5	$CB_G o_1 = 1$
BBL3	189	756	4	3.485	1067	42.853	45.841	5	$CB_G(o_1 = 1 \wedge o_2 = 2)$
BBL4	1595	6380	0	0	10295	0.486	6.05	9	$EB_G(o_1 = 1 \wedge o_2 = 2 \wedge o_3 = 3)$
BBL5	1595	6380	0	0	10295	0.39	5.007	9	$EB_G o_1 < o_2$
BBL6	2	8	0	0	11	0.663	0.009	1	$DB_G o_1 < o_2$

Table 1: Result for three domains (N0-N6, G0-G6, BBL0-BBL6 are instances for Number, Grapevine and BBL respectively). G represents the group of agents – $\{a, b\}$ for Number and BBL; and, $\{a, b, c, d\}$ for Grapevine. 'Exp" and "Gen" are the number of nodes expanded and generated during search, "Max" and "Avg" under "Common" as the maximum and average level of nesting required to compute cf_G, $|calls|$ and "Avg Time(ms)" under "External" as the number and average time of external function calls, and $|p|$ as plan length. Since we implement the ternary semantics, we denote the ternary evaluation result $T[\bar{s}, \varphi]$ equal to 0, $\frac{1}{2}$ and 1 as $\neg\varphi$, $\varphi \rightarrow \frac{1}{2}$ and φ respectively.

Different from the plan of items (G1) and (G5), item (G6) contains $B_a EB_G sct_a \rightarrow \frac{1}{2}$. So that a's secret is shared by another agent to b in another room rather than a.

D .3 BBL

In those instance, we limit the camera's angle of turning to a set $\{0°, \pm 45°, \pm 90°, \pm 135°, 180°\}$ and the field is in a 5×5 grid for simplicity. Initially, camera a and b locates at $(3, 3)$ and $(1, 1)$ with direction of $-135°$ and $90°$, while o_1, o_2 and o_3 locates at $(0, 0)$, $(2, 2)$ and $(3, 3)$, with value 1, 2 and 3 respectively.

Items (BBL5) and (BBL6) are worthy of discussion. For item (BBL5), both agents need to turn around to see both o_1 and o_3. While for item (BBL6), only agent b turns to see o_3. Agent a and b's individual perspectives are $[\{o_1 = 1\}, \{o_1 = 1\}]$ and $[\{\}, \{o_3 = 3\}]$ respectively. By forming df_G, which is $\{[\{o_1 = 1\}, \{o_1 = 1, o_3 = 3\}]\}$ (same perspective for both a and b), we have $DB_G o_1 < o_3$.

E Distributed Perspective Examples

Here, we provide visualization of those three distributed perspectives examples in Example 2.

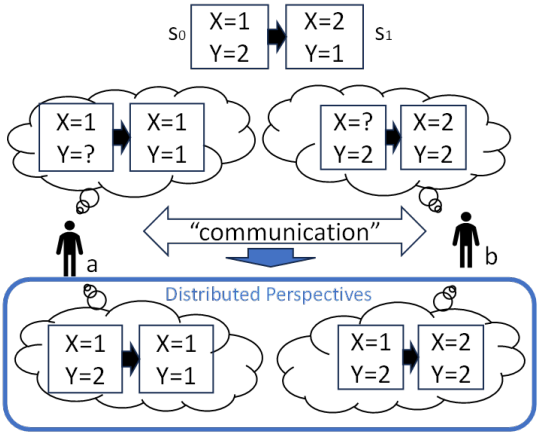

Figure 1: Example 2.1

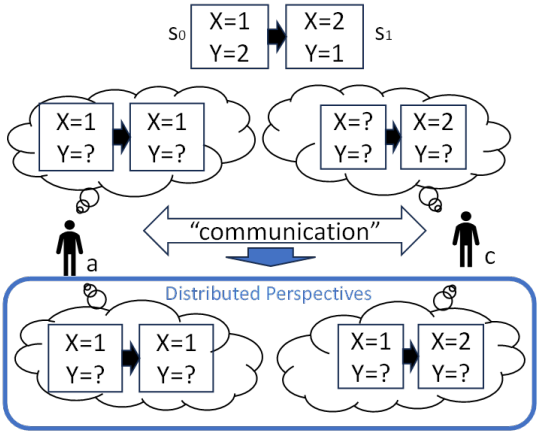

Figure 2: Example 2.3

References

[Hu *et al.*, 2022] Guang Hu, Tim Miller, and Nir Lipovetzky. Planning with perspectives – decomposing epistemic planning using functional STRIPS. *JAIR*, 75:489–539, October 2022.

[Hu *et al.*, 2023] Guang Hu, Tim Miller, and Nir Lipovetzky. Planning with multi-agent belief using justified perspectives. *Proceedings of the International Conference on Automated Planning and Scheduling*, 33(1):180–188, Jul. 2023.

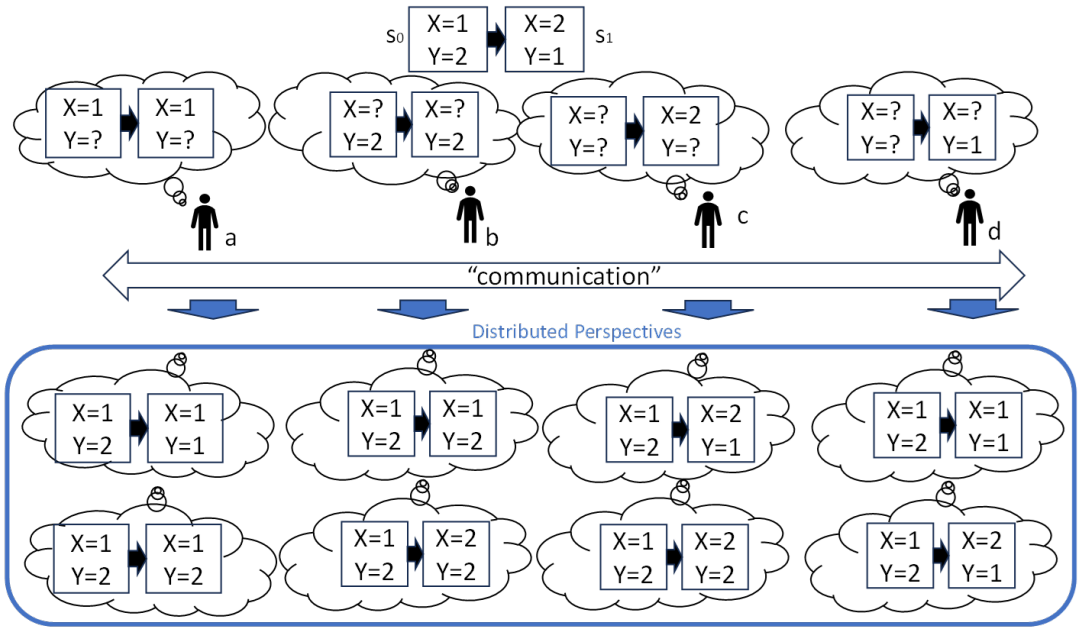

Figure 3: Example 2.2